Stat 6410 (Ghosh) Midterm 2: Solutions

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Numbered results refer to the course text.

Q1

Let m be the Lebesgue measure on $[0,1]$ and $\mathbf{1}_A(x)$ denote the indicator function of A. Define a sequence of measurable transformations on $(\Omega = [0,1], \mathcal{F} = \mathcal{B}[0,1], \mu = m)$: For $n \geq 1$, and $x \in \Omega = [0,1]$,

$$f_n(x) = n\mathbf{1}_{[\frac{1}{n}, \frac{2}{n}]}(x) = \begin{cases} n & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 0 & \text{otherwise} \end{cases}$$

The limit function is $f(x) \equiv 0$ for all $x \in \Omega = [0,1]$.

(a)

Show that $f_n \to f$ almost everywhere (m).

Answer

We show that $f_n(x) \to f(x)$ for every $x \in \Omega$. Let $x \in (0,1]$ be given. For every $n > 2/x$ we have $\frac{2}{n} < x$, so $x \notin [\frac{1}{n}, \frac{2}{n}]$ and $f_n(x) = 0$; hence $f_n(x) = 0$ for all but finitely many n, and $f_n(x) \to 0 = f(x)$. (For $x = 0$: since $0 < \frac{1}{n}$ for every n, $f_n(0) = 0$ for all $n \geq 1$.) Thus $f_n \to f$ pointwise on Ω (Definition 2.5.1), so Definition 2.5.2 holds with $B = \emptyset$, and $f_n \to f$ a.e. (m). //

(b)

Does the sequence $f_n \to f$ in measure?

Answer

Yes, $f_n \to f$ in measure. Since $m(\Omega) = 1 < \infty$ and $f_n \to f$ a.e. (m) by (a), Theorem 2.5.1 gives $f_n \xrightarrow{m} f$. (Directly: for each $\epsilon > 0$, $\{|f_n - f| > \epsilon\} \subset [\frac{1}{n}, \frac{2}{n}] \cap [0,1]$, so $m(\{|f_n - f| > \epsilon\}) \leq \frac{1}{n} \to 0$, as required by Definition 2.5.3.) //

(c)

For what values of $0 < p < \infty$ does the sequence $f_n \to f$ in L_p?

Answer

$f_n \to f$ in L^p if and only if $0 < p < 1$. Each f_n is bounded and $m(\Omega) = 1$, so $f_n \in L^p$; also $f \equiv 0 \in L^p$. For $n \geq 2$, $[\frac{1}{n}, \frac{2}{n}] \subset [0,1]$ and $|f_n - f|^p = n^p \mathbf{1}_{[1/n,\, 2/n]}$ is simple, so by Definition 2.3.2

$$\int |f_n - f|^p \, dm = n^p \, m\big([\tfrac{1}{n}, \tfrac{2}{n}]\big) = n^{p-1}.$$

(For $n = 1$, $[1,2] \cap [0,1] = \{1\}$, so $f_1 = \mathbf{1}_{\{1\}}$ and the integral is 0; a single term does not affect the limit.) Hence $\int |f_n - f|^p \, dm \to 0$ if $0 < p < 1$, whereas it equals 1 for all $n \geq 2$ if $p = 1$ and diverges to ∞ if $p > 1$. By Definition 2.5.4, $f_n \to f$ in L^p exactly when $0 < p < 1$. //

(d)

Is this sequence $\{f_n\}$ uniformly integrable?

Answer

No, $\{f_n\}$ is not UI. By Definition 2.5.8, UI requires $\sup_{n \geq 1} a_n(t) \to 0$ as $t \to \infty$, where $a_n(t) \equiv \int_{\{|f_n| > t\}} |f_n| \, dm$. Let $t > 0$ be given, and take any integer $n > \max\{t, 1\}$. Then $\{|f_n| > t\} = [\frac{1}{n}, \frac{2}{n}]$, so

$$\sup_{k \geq 1} a_k(t) \geq a_n(t) = \int_{[1/n,\, 2/n]} n \, dm = 1.$$

Since t was arbitrary, $\sup_{n \geq 1} a_n(t) \not\to 0$ as $t \to \infty$. (Alternatively: if $\{f_n\}$ were UI, then since $m(\Omega) < \infty$ and $f_n \to f$ a.e. (m), Theorem 2.5.10 would give $\int |f_n - f| \, dm \to 0$, contradicting (c) with $p = 1$.) //

(e)

Consider the sequence $\{g_n\}$ where $g_n = \cos(f_n)$ (here $\cos(x) = \text{cosine}(x)$ is the well-known trigonometric function). Does this sequence $\{g_n\}$ converge to $g \equiv 1$ almost surely (m)?

Answer

Yes, $g_n \to g$ at every point of Ω, hence a.s. (m). Explicitly,

$$g_n(x) = \cos\big(f_n(x)\big) = \begin{cases} \cos n & \text{if } \frac{1}{n} \leq x \leq \frac{2}{n} \\ 1 & \text{otherwise,} \end{cases}$$

and each g_n is Borel measurable since cos is continuous (Propositions 2.1.2 and 2.1.1(ii)). Let $x \in [0,1]$ be given. By (a), $f_n(x) = 0$ for all sufficiently large n (all n if $x = 0$; all $n > 2/x$ if $x > 0$), so $g_n(x) = \cos 0 = 1$ for all such n, and $g_n(x) \to 1$. Since x was arbitrary, $g_n \to g$ everywhere on Ω; in particular $g_n \to g$ a.s. (m), with exceptional set $\emptyset$. //

Q2

You can state & use any pre-measure theory facts you know (if needed), but justify all the measure theory steps.

(i)

Show that a Binomial($n = 10, p = 0.1$) random variable exists: In other words, using steps of the Caratheodory extension theorem, exhibit a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \to \mathbb{R}$, such that its p.m.f. is:

$$P(X = k) = f(k) = \binom{10}{k}(0.1)^k(0.9)^{10-k}, \quad k \in \{0, 1, \ldots, 10\},$$

and $P(X = k) = f(k) = 0$, otherwise.

Answer

Sketch. Take P to be the Lebesgue–Stieltjes measure of the Binomial$(10, 0.1)$ cdf, obtained by extending a measure on the semialgebra of half-open intervals via Carathéodory's theorem, and take X to be the identity map, as in Section 2.2 following Definition 2.2.4.

Proof. Let $S \equiv \{0, 1, \ldots, 10\}$ and

$$F(x) \equiv \sum_{k \in S,\, k \le x} f(k), \quad x \in \mathbb{R}.$$

Since $f \ge 0$, F is nondecreasing. It is constant on $(-\infty, 0)$, on each $[j, j+1)$ for $j = 0, \ldots, 9$, and on $[10, \infty)$, so it is right continuous, with jumps $F(x) - F(x-) = f(x)$ for $x \in S$ and 0 for $x \notin S$. Also $F(-\infty) = 0$ and, by the binomial theorem, $F(\infty) = \sum_{k=0}^{10} \binom{10}{k}(0.1)^k(0.9)^{10-k} = (0.1 + 0.9)^{10} = 1$. So F is a cdf (Definition 2.2.4).

Let $\mathcal{C} \equiv \{(a, b] \cap \mathbb{R} : -\infty \le a \le b \le \infty\}$, a semialgebra on $\mathbb{R}$ with $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$, and define

$$\mu_F\big((a, b] \cap \mathbb{R}\big) \equiv F(b) - F(a).$$

Then $\mu_F \ge 0$, $\mu_F(\emptyset) = 0$, and μ_F is finitely additive on $\mathcal{C}$ by telescoping. It is also countably additive on $\mathcal{C}$. Let disjoint $C_i \in \mathcal{C}$ with $\bigcup_{i \ge 1} C_i = C \in \mathcal{C}$ be given. Finite additivity and monotonicity give $\sum_i \mu_F(C_i) \le \mu_F(C)$. For the reverse (say $C = (a, b]$ bounded; the unbounded case follows by letting the endpoints tend to $\pm\infty$), fix $\epsilon > 0$. Right continuity of F gives $a' \in (a, b)$ and $b_i' > b_i$ with $F(a') - F(a) < \epsilon$ and $F(b_i') - F(b_i) < \epsilon 2^{-i}$, where $C_i = (a_i, b_i]$. By Heine–Borel, finitely many of the open (a_i, b_i') cover the compact $[a', b]$, so $F(b) - F(a') \le \sum_i [F(b_i') - F(a_i)]$ and hence $\mu_F(C) \le \sum_i \mu_F(C_i) + 2\epsilon$ (cf. Problems 1.22–1.23).

By Carathéodory's extension theorem, the outer measure

$$\mu_F^*(A) \equiv \inf\Big\{ \sum_{n\geq 1} \mu_F(C_n) : C_n \in \mathcal{C},\ A \subset \bigcup_{n\geq 1} C_n \Big\}, \quad A \subset \mathbb{R},$$

is a measure on the σ-algebra $\mathcal{M}_{\mu_F^*}$ of sets A satisfying $\mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^c)$ for all $E \subset \mathbb{R}$; moreover $\mathcal{C} \subset \mathcal{M}_{\mu_F^*}$ and $\mu_F^* = \mu_F$ on $\mathcal{C}$. Since $\mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{C}\rangle \subset \mathcal{M}_{\mu_F^*}$, we set $P \equiv \mu_F^*|_{\mathcal{B}(\mathbb{R})}$, a measure with $P(\mathbb{R}) = F(\infty) - F(-\infty) = 1$, and unique as an extension of μ_F since μ_F is finite. This is the Lebesgue–Stieltjes measure of F (Theorem 1.3.3, Definition 1.3.7).

Now let $(\Omega, \mathcal{F}, P) \equiv (\mathbb{R}, \mathcal{B}(\mathbb{R}), P)$ and $X(\omega) \equiv \omega$. For each $a \in \mathbb{R}$, $X^{-1}((-\infty, a]) = (-\infty, a] \in \mathcal{B}(\mathbb{R})$, so X is a random variable (Definition 2.1.2(b)). Let $x \in \mathbb{R}$ be given. Since $(x - \frac{1}{n}, x] \downarrow \{x\}$ and P is finite, m.c.f.a. (Proposition 1.2.3) gives

$$P(X = x) = P(\{x\}) = \lim_{n\to\infty} \Big[F(x) - F\big(x - \tfrac{1}{n}\big)\Big] = F(x) - F(x-) = \begin{cases} f(x) & \text{if } x \in S, \\ 0 & \text{if } x \notin S. \end{cases}$$

Since x was arbitrary, X has the Binomial$(10, 0.1)$ p.m.f. //

(ii)

Compute the following Lebesgue integrals (give a numerical answer)

$$\int_\Omega X^2(\omega)\, dP(\omega)$$

Answer

$\int_\Omega X^2\, dP = 1.9$. Since $x \mapsto x^2$ is continuous, X^2 is a nonnegative $\langle\mathcal{F}, \mathcal{B}(\mathbb{R})\rangle$-measurable function (Propositions 2.1.2 and 2.1.1(ii)), so $\int_\Omega X^2\, dP \in [0, \infty]$ is defined (Definition 2.3.3). For $k \in S$ let $A_k \equiv X^{-1}(\{k\}) \in \mathcal{F}$ (in the construction of (i), $A_k = \{k\}$). The A_k are disjoint with $P\big(\bigcup_{k\in S} A_k\big) = \sum_{k\in S} f(k) = 1$, and $X^2 = s$ on $\bigcup_{k\in S} A_k$, where

$$s \equiv \sum_{k=0}^{10} k^2\, \mathbf{1}_{A_k}$$

is a nonnegative simple function (Definition 2.3.1). Hence $X^2 = s$ a.e. (P), and by Corollary 2.3.3 (with Proposition 2.3.1(iii)) and Definition 2.3.2,

$$\int_\Omega X^2\, dP = \int_\Omega s\, dP = \sum_{k=0}^{10} k^2 P(A_k) = \sum_{k=0}^{10} k^2 \binom{10}{k} (0.1)^k (0.9)^{10-k}.$$

For the finite sum (a pre-measure theory fact), write $k^2 = k(k-1) + k$ and use $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$, $k\binom{n}{k} = n\binom{n-1}{k-1}$, and the binomial theorem:

$$\sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k} = n(n-1)p^2 + np = np(1-p) + (np)^2.$$

With $n = 10$ and $p = 0.1$, this is $0.9 + 1 = 1.9$. //

Q3

Answer the following:

(a)

For a *nonnegative* random variable X, prove $\sqrt{1+(EX)^2} \le E\sqrt{1+X^2} \le 1+EX$, considering the possibility that $E(X)$ can be infinite.

Answer

Let $\phi(x) \equiv \sqrt{1+x^2}$, $x \in \mathbb{R}$. Since $\phi''(x) = (1+x^2)^{-3/2} > 0$, ϕ is convex on $\mathbb{R}$ (Proposition 3.1.8). As ϕ is continuous and $X \ge 0$, $\phi(X)$ is nonnegative and measurable, so $E\phi(X) \in [0,\infty]$ is defined. For $x \ge 0$, $x^2 \le 1+x^2 \le (1+x)^2$, so

$$x \le \sqrt{1+x^2} \le 1+x, \qquad x \ge 0. \tag{$*$}$$

Case 1: $EX < \infty$. By the right inequality in $(*)$, and monotonicity and linearity of the integral (Corollary 2.3.3),

$$E\sqrt{1+X^2} \le E(1+X) = 1+EX < \infty.$$

Hence $E|X| = EX < \infty$ and $E|\phi(X)| < \infty$, and $P(X \in (-\infty,\infty)) = 1$, so Jensen's inequality (Theorem 3.1.9) gives $\phi(EX) \le E\phi(X)$, i.e., $\sqrt{1+(EX)^2} \le E\sqrt{1+X^2}$.

Case 2: $EX = \infty$. Then $1+EX = \infty$ and $\sqrt{1+(EX)^2} = \infty$. By the left inequality in $(*)$ and monotonicity, $E\sqrt{1+X^2} \ge EX = \infty$, so all three quantities equal ∞.

In either case, $\sqrt{1+(EX)^2} \le E\sqrt{1+X^2} \le 1+EX$. //

(b)

Let $(\Omega = \mathbb{R}, \mathcal{F} = \mathcal{B}(\mathbb{R}))$. Let m be the Lebesgue measure and μ be the standard normal (Gaussian) measure. That is,

$$\mu(A) = \int_A \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\, dm(x), \quad A \in \mathcal{B}(\mathbb{R})$$

Show that the RN derivative of Lebesgue measure m w.r.t. μ to be:

$$\left(\frac{dm}{d\mu}\right)(x) = \sqrt{2\pi}\, e^{\frac{x^2}{2}}, \quad x \in \mathbb{R}.$$

(you can state any result proved in class to argue this)

Answer

Let $g(x) \equiv \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and $h(x) \equiv \sqrt{2\pi}\, e^{x^2/2} = 1/g(x)$; both are continuous, hence Borel, and strictly positive. By definition $\mu(A) = \int_A g\, dm$ for all $A \in \mathcal{B}(\mathbb{R})$, so $\mu \ll m$ (Example 4.1.1) with $\frac{d\mu}{dm} = g$ (Definition 4.1.3). Since m and μ are σ-finite and $\frac{d\mu}{dm} = g > 0$ everywhere, Proposition 4.1.2(iii) gives $m \ll \mu$ and

$$\frac{dm}{d\mu} = \left(\frac{d\mu}{dm}\right)^{-1} = \frac{1}{g} = \sqrt{2\pi}\, e^{x^2/2} \quad \text{a.e. } (\mu).$$

To check this directly, note that $\int u\, d\mu = \int ug\, dm$ for every Borel $u : \mathbb{R} \to [0, \infty]$: it holds for indicators by the definition of μ, for nonnegative simple u by linearity (Proposition 2.3.1(i)), and for general u by the MCT (Theorem 2.3.4) applied to both sides along nonnegative simple $u_n \uparrow u$, since then $u_n g \uparrow ug$ (cf. Problem 2.39). Taking $u = h\mathbf{1}_A$ and using $hg \equiv 1$,

$$\int_A h\, d\mu = \int \mathbf{1}_A\, hg\, dm = m(A), \quad A \in \mathcal{B}(\mathbb{R}),$$

so h is a version of $\frac{dm}{d\mu}$ (Definition 4.1.3), which is unique up to μ-null sets. //

(c)

Let $(\Omega, \mathcal{F})$ be a measurable space with two measures μ and ν on it. Given,

$$\mu \ll \nu, \quad \text{and} \quad \mu \perp \nu.$$

Show that $\mu(A) = 0$ for all $A \in \mathcal{F}$.

Answer

Since $\mu \perp \nu$, there is $B \in \mathcal{F}$ with $\mu(B) = 0$ and $\nu(B^c) = 0$ (Definition 4.1.2). Let $A \in \mathcal{F}$ be given. By monotonicity, $\mu(A \cap B) \leq \mu(B) = 0$ and $\nu(A \cap B^c) \leq \nu(B^c) = 0$, and the latter gives $\mu(A \cap B^c) = 0$ since $\mu \ll \nu$ (Definition 4.1.1). Hence

$$\mu(A) = \mu(A \cap B) + \mu(A \cap B^c) = 0.$$

Since A was arbitrary, $\mu(A) = 0$ for all $A \in \mathcal{F}$. //